



Figure 1: Centralised secrets manager

Common integration pitfalls

When organisations fail to integrate these concepts properly, they create dangerous gaps:

- **The ‘plaintext leak’ in logs:** An application receives data securely via TLS 1.3, but before encrypting it to the database, an internal error occurs. The application accidentally writes the raw data to a debugging log file on an unencrypted disk drive.
- **Unencrypted local backups:** A system perfectly encrypts data on its live database (at rest) and transfers data securely via API (in transit). However, night-time database backups are saved as raw `.sql` files and compressed into a zip folder without encryption.
- **Decrypting at the wrong boundary:** Decrypting TLS traffic at a cloud load balancer and then sending that traffic completely unencrypted across internal backend networks to the database server. To fix this, modern systems use mTLS (mutual TLS) to ensure data remains encrypted in transit even inside the internal corporate network.

Here’s the action that should be taken w.r.t. encryption.

Individuals should:

- Turn on full-disk encryption via VeraCrypt or LUKS.
- Use Cryptomator for files stored in OneDrive, iCloud, or Google Drive.
- Keep passwords secure using an open source password manager like Bitwarden.

Teams and organisations can:

- Enforce TLS 1.3 across all public web properties and disable TLS 1.0/1.1.
- Automate certificate renewals using Let’s Encrypt and cert-manager.
- Use WireGuard for all remote company access.
- Secure secrets and encryption keys using HashiCorp Vault.

Why open source matters for security

You might wonder: “If the code is public, can’t attackers find weaknesses more easily?”

The opposite is true. When code is open source, thousands of security experts can inspect it, find bugs, and fix them. With closed source software, you’re trusting one company to get it right—and you can’t verify their claims.

Tools like VeraCrypt, LUKS, and Let’s Encrypt have been reviewed by security professionals worldwide. Their transparency is their strength.

Encryption isn’t just for spies and banks. In 2026, it’s a necessity for anyone who values everyone. Start small. Encrypt your laptop today. Set up HTTPS on your website this week. Every step you take makes you—and the people who trust you with their data—safer.

The lock on your digital door is free. There’s no excuse not to use it. **END** 

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